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```
#sales data
import pandas as pd
sales_data = {
    'product' : ['Laptop', 'Mouse', 'Keyboard', 'Monitor', 'Laptop', 'Mouse', 'Keyboard', 'Monitor'],
    'Region' : ['North', 'North', 'South', 'South', 'North', 'South', 'North', 'South'],
    'Quantity' : [5, 15, 12, 8, 7, 10, 6, 9],
    'Price' : [1000, 25, 75, 300, 1000, 25, 75, 300]}
sales = pd.DataFrame(sales_data)
print("====Sales data====")
print(sales)
'''
Imported Pandas, stored multiple sales records in a dictionary (sales_data),
converted it into a DataFrame (sales), and displayed the sales dataset.
'''

#Exercise 1: Calculate total revenue for each row

sales['Revenue']=sales['Quantity']*sales['Price']
print("\n =====With Revenue Column=====")
print(sales)
print(f"\n Total Revenue: ${sales['Revenue'].sum():,.2f}")

'''
Revenue column values are summed to calculate and display the overall total revenue in a
formatted currency format.
'''

#Exercise 2: Revenue by Product

print("\n =====Revenue by product=====")
product_revenue = sales.groupby('product')['Revenue'].sum().sort_values(ascending=False)
print(product_revenue)

'''
Groups the data by product, calculates the total revenue for each product, sorts the results in descending order,
and displays the revenue generated by each product.
'''
```

#Exercise 3: Revenue by Region

```
print("\n====Revenue by Region====")
region_revenue = sales.groupby('Region')['Revenue'].sum()
print(region_revenue)
print(f"\nHighest Revenue Region: {region_revenue.idxmax()} with ${region_revenue.max():,.2f}")

'''
Finds the region with the highest total revenue and displays its name along with the revenue
amount in a formatted currency format.
'''
```

#Exercise 4: Complex aggregation

```
print("\nDetailed Analysis by Region and Product:")
analysis = sales.groupby(['Region', 'product']).agg({
    'Quantity': 'sum',
    'Revenue': 'sum',
    'Price': 'first'}).reset_index()
print(analysis)
print("\n Sorted by Revenue (highest first):")
print(analysis.sort_values('Revenue', ascending=False))

'''
Groups sales data by Region and Product, calculates total Quantity and Revenue,
retains the Product Price, and displays the results sorted by highest revenue.
'''
```

```
=====Sales data=====
```

	product	Region	Quantity	Price
0	Laptop	North	5	1000
1	Mouse	North	15	25
2	Keyboard	South	12	75
3	Monitor	South	8	300
4	Laptop	North	7	1000
5	Mouse	South	10	25
6	Keyboard	North	6	75
7	Monitor	South	9	300

```
=====With Revenue Column=====
```

	product	Region	Quantity	Price	Revenue
0	Laptop	North	5	1000	5000
1	Mouse	North	15	25	375
2	Keyboard	South	12	75	900
3	Monitor	South	8	300	2400
4	Laptop	North	7	1000	7000
5	Mouse	South	10	25	250
6	Keyboard	North	6	75	450
7	Monitor	South	9	300	2700

Total Revenue: \$19,075.00

```
=====Revenue by product=====
```

product

Laptop 12000

Monitor 5100

Keyboard 1350

Mouse 625

Name: Revenue, dtype: int64

```
=====Revenue by Region=====
```

```
Region
```

```
North      12825
```

```
South      6250
```

```
Name: Revenue, dtype: int64
```

```
Highest Revenue Region: North with $12,825.00
```

```
Detailed Analysis by Region and Product:
```

	Region	product	Quantity	Revenue	Price
0	North	Keyboard	6	450	75
1	North	Laptop	12	12000	1000
2	North	Mouse	15	375	25
3	South	Keyboard	12	900	75
4	South	Monitor	17	5100	300
5	South	Mouse	10	250	25

```
Sorted by Revenue (highest first):
```

	Region	product	Quantity	Revenue	Price
1	North	Laptop	12	12000	1000
4	South	Monitor	17	5100	300
3	South	Keyboard	12	900	75
0	North	Keyboard	6	450	75
2	North	Mouse	15	375	25
5	South	Mouse	10	250	25